

Lecture 13: Speaker Notes

Introduction to ADNI and Modeling MCI Progression

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1 Overview

Lecture Duration: 80 minutes **Format:** Presentation with discussion **Goal:** Introduce ADNI dataset and establish Project 2 hypothesis (Grassi et al. replication)

1.1 Prerequisites

Before class, verify:

- ☐ Students have completed Lecture 12 (renv + Docker setup)

- ☐ ADNI Data Use Agreement process explained
- ☐ Grassi et al. (2019) PDF posted to Canvas
- ☐ Donohue et al. (2026) PDF posted to Canvas
- ☐ table2.png displays correctly in slides

1.2 Materials Needed

- Slides (slides.pdf or slides.html)
- Research papers: Grassi (2019), Rye (2022), Donohue (2026)
- table2.png (Grassi Table 2 for visual reference)
- ADNI website open: adni.loni.usc.edu

2 Timing Plan

Time	Cumulative	Section	Notes
0:00	0 min	Title, Course Info	Quick start
0:02	2 min	Today's Topics	Overview of 7 topics
0:05	5 min	Part I: ADNI Study	History and design
0:15	20 min	Part II: Understanding MCI	Clinical context
0:10	30 min	Part III: Project Hypothesis	Grassi approach
0:15	45 min	Part IV: Key Predictors	Feature categories
0:10	55 min	Part V: ML Methods	Algorithm overview
0:15	70 min	Part VI: Project Assignment	Requirements
0:10	80 min	Summary, Q&A	Wrap-up

3 Part I: The ADNI Study (15 min)

3.1 What is ADNI? (Slide 8)

Key points:

- Emphasize ADNI as the “gold standard” for AD research data
- Started 2004 under Michael Weiner (UCSF) and Leon Thal (UCSD)
- Public-private partnership model
- Data freely available (with DUA)

Discussion prompt: “How many of you have heard of ADNI before?”

3.2 ADNI Study Design (Slide 9)

Key points:

- Walk through each phase and its focus
- Highlight the evolution: ADNI-1 established methods, later phases added biomarkers
- ADNI-4 focuses on diversity and novel digital biomarkers
- Total N = 3,032 is substantial for AD research

Anticipated question: “Why so many phases?” **Answer:** Science evolves; new biomarkers (tau PET) and populations (diverse recruitment) require new cohorts.

3.3 Diagnostic Groups (Slide 10)

Key points:

- CN = Cognitively Normal (healthy controls)
- EMCI vs LMCI distinction based on Logical Memory II scores
- Emphasize that ADNI uses specific, operationalized criteria

Teaching tip: Draw the progression on board: CN → EMCI → LMCI → Dementia

3.4 The alzverse (Slides 11-12)

Key points:

- NEW in 2026: ADNIMERGE2 replaces original ADNIMERGE
- Emphasize reproducibility benefits: vignettes, ADaM format
- Show GitHub documentation links
- Mention ~250 citations for original package

Anticipated question: “Can I still use the old ADNIMERGE?” **Answer:** Yes, but ADNIMERGE2 has updated data and better documentation. Recommend using ADNIMERGE2 for new projects.

3.5 ADNI Baseline Characteristics (Slide 14)

Key points:

- 44% MCI at baseline - this is our target population
- Note diversity improvements in recent phases (ADNI-4: 35% Black/AA)
- Mean age ~72 years, well-educated sample (16 years)

Discussion: “What are limitations of this sample for generalizability?”

4 Part II: Understanding MCI (10 min)

4.1 MCI Definition (Slide 17)

Key points:

- Petersen criteria are the standard
- Emphasize: NOT demented, but NOT normal
- Subjective complaint + objective impairment
- ADLs preserved (distinguishes from dementia)

Teaching tip: Ask students to articulate why each criterion matters.

4.2 MCI Subtypes (Slide 18)

Key points:

- Amnestic vs non-amnestic: memory primary or not
- Single vs multi-domain: how many cognitive areas affected
- ADNI focuses on amnestic MCI (highest AD risk)

4.3 The Clinical Problem (Slide 19)

Key points:

- THIS IS THE MOTIVATION for the project
- Not all MCI converts - only 10-15% annually
- Clinical need: who will progress vs remain stable?
- Applications: treatment planning, trial enrollment, counseling

Anticipated question: “Why not just wait and see?” **Answer:** Early intervention may be more effective; patients/families want to plan; clinical trials need enriched populations.

4.4 MCI to AD Conversion Rates (Slide 20)

Key points:

- From Grassi: 36% converted over 3 years
- This is the outcome students will predict
- Emphasize binary classification: converter vs stable

5 Part III: Project 2 Hypothesis (10 min)

5.1 Grassi et al. Study (Slide 22)

Key points:

- This paper defines the student project
- “Clinically translatable” = uses only standard clinical measures
- No imaging, no genetics (intentionally)
- Goal: practical tool for community clinics

Read the abstract aloud or have students read key sentences.

5.2 Study Design (Slide 23)

Key points:

- $N = 550$ MCI subjects
- Baseline only (no longitudinal features)
- 3-year follow-up window
- Site-independent split is critical for generalization

5.3 Key Results (Slide 24)

Key points:

- AUROC 0.88 is excellent for clinical prediction
- Sensitivity 77.7%: catches most converters
- Specificity 79.9%: few false alarms
- PPV/NPV are clinically meaningful

Teaching tip: Have students interpret: “If you test positive, what’s the probability you’ll actually convert?” (PPV = 71.9%)

5.4 Table 2 Image (Slide 25)

Key points:

- Show the actual table from the paper
- This is what students should reproduce
- Compare converters vs stable on each variable
- Note which variables differ significantly

5.5 Why This Approach? (Slide 26)

Key points:

- Clinical translatability is the key innovation
- Contrast with high-tech approaches (PET, CSF, genetics)
- Community clinics can use these measures
- Trade accuracy for accessibility

6 Part IV: Key Predictors (15 min)

6.1 Feature Categories (Slide 28)

Key points:

- Four categories: sociodemographic, clinical scales, neuropsych, MCI subtype
- Note what's EXCLUDED: imaging, genetics
- This is a deliberate choice for clinical applicability

6.2 Sociodemographic Features (Slide 29)

Key points:

- Age, education, sex, marital status
- These are “free” - already collected in clinic
- Marital status may proxy for social support/cognitive reserve

6.3 Clinical Scales (Slide 30)

Key points:

- CDR-SB: Clinical Dementia Rating - Sum of Boxes
 - Clinician-rated based on interview
 - 6 domains, 0-18 scale
- FAQ: Functional Activities Questionnaire
 - Informant-rated
 - Instrumental ADLs

Teaching tip: Show example CDR-SB items if time permits.

6.4 Neuropsychological Tests (Slide 31)

Key points:

- Walk through each test briefly
- MMSE: screening, ceiling effects in MCI
- ADAS-Cog: more detailed, designed for AD trials
- RAVLT: verbal memory, very sensitive to AD
- LDELTOTAL: Logical Memory delayed recall
- TMTB: executive function, processing speed

6.5 ADAS-Cog Detail (Slide 32)

Key points:

- ADAS11 vs ADAS13: 13 adds delayed recall + cancellation
- ADAS13 is more sensitive to change
- Higher scores = worse performance (counterintuitive)

6.6 RAVLT Detail (Slide 33)

Key points:

- 15-word list, 5 learning trials
- Interference list, then recall
- 30-minute delayed recall
- Derived scores: immediate, learning, forgetting

Teaching tip: Demonstrate briefly: “I’ll read 15 words, you try to remember as many as possible...”

6.7 Feature Importance (Slide 34)

Key points:

- ADAS13 most important overall
- CDR-SB and FAQ: clinical staging
- RAVLT forgetting: memory retention is key
- MMSE less important than expected (ceiling effects)

Discussion: “Why might memory retention be more predictive than learning?”

7 Part V: Machine Learning Methods (10 min)

7.1 Overview (Slide 36)

Key points:

- Ensemble of 13 algorithms
- No single best algorithm
- Weighted rank averaging for final prediction

7.2 Individual Algorithms (Slide 37)

Key points:

- Walk through categories briefly
- Students don’t need to implement all 13
- Project requires at least 3 different algorithms

7.3 Random Forest Example (Slide 38)

Key points:

- Show code structure
- ntree, mtry parameters
- Variable importance output

Note: Students can use caret or tidymodels to simplify this.

7.4 Cross-Validation Strategy (Slide 39)

Key points:

- Site-independent split is KEY
- This prevents “cheating” on generalization
- 5-fold CV within training set
- Held-out test set for final evaluation

Anticipated question: “Why site-independent?” **Answer:** Sites have systematic differences (equipment, protocols, patient populations). Testing on new sites simulates real deployment.

7.5 Evaluation Metrics (Slides 40-41)

Key points:

- Define each metric clearly
- AUROC: overall discrimination
- Sensitivity/specificity: threshold-dependent
- ROC curve shows trade-off

8 Part VI: Project Assignment (15 min)

8.1 Project Overview (Slide 43)

Key points:

- Read requirements clearly
- Due March 3, 2026
- Four deliverables: SAP, code, report, presentation

8.2 Required Components (Slide 44)

Key points:

- Data preparation: cohort selection, feature extraction
- Model development: at least 3 algorithms
- Evaluation: ROC, sensitivity/specificity, importance

8.3 Data Preparation Code (Slides 45-46)

Key points:

- Walk through code step by step
- ADSL for subject-level data
- ADQS for cognitive outcomes
- Show how to define converter status

Teaching tip: Have students try this code in class if time permits.

8.4 Timeline (Slide 47)

Key points:

- 3 weeks total
- Week 1: data prep
- Week 2: modeling
- Week 3: evaluation and report

Emphasize: Start early! Data cleaning takes time.

8.5 Resources (Slide 48)

Key points:

- Papers on Canvas
- ADNIMERGE2 from ADNI website
- alzverse documentation online
- Office hours available

9 Summary and Q&A (10 min)

9.1 Summary (Slide 49)

Review key points:

1. ADNI: major resource, N=3,032, public data
2. MCI: transitional state, 36% convert in 3 years
3. Grassi approach: clinical measures only, AUROC=0.88
4. Your project: replicate with ADNIMERGE2

9.2 Next Steps (Slide 50)

Action items:

1. Apply for ADNI access TODAY
2. Download ADNIMERGE2 when approved
3. Read Grassi paper thoroughly
4. Start data exploration
5. Draft SAP

10 Troubleshooting Guide

10.1 Common Issues

“ADNI access is taking too long”

- DUA approval can take 1-2 weeks
- Students should apply immediately
- Instructor can provide subset for initial exploration

“ADNIMERGE2 won’t install”

- Check R version (needs 4.0+)
- Install dependencies first
- Try: `install.packages("ADNIMERGE2_x.x.x.tar.gz", repos = NULL, type = "source")`

“Can’t find MCI subjects”

- Filter on DX_BL (baseline diagnosis)
- Use: `filter(DX_BL %in% c("EMCI", "LMCI"))`

“How do I define converters?”

- Any subsequent diagnosis of “Dementia”
- Use longitudinal DX variable, not just baseline

“My AUROC is much lower than Grassi”

- Check feature engineering
- Verify train/test split
- Consider different algorithms
- 0.70-0.80 is still acceptable for student project

11 Anticipated Questions

Q: Do we need to use all 13 algorithms like Grassi?

A: No. Minimum 3 different algorithms required. Focus on understanding the methods rather than replicating everything.

Q: Can we include imaging or genetic features?

A: For the core project, follow Grassi (clinical only). You may explore imaging/genetics as an extension if time permits.

Q: What if my results differ from Grassi?

A: Expected! Different random seeds, slightly different cohort selection, different software versions. Focus on methodology, not exact replication.

Q: How do I handle missing data?

A: Document your approach. Options: complete case analysis, imputation, or exclude variables with high missingness. Justify your choice.

Q: Can we work in groups?

A: Analysis should be individual. You may discuss concepts and debug code together, but each student submits their own work.

12 Post-Lecture Checklist

- ☐ Remind students to apply for ADNI access today
- ☐ Post Donohue et al. (2026) to Canvas
- ☐ Update Canvas assignment with due date
- ☐ Schedule office hours for data questions
- ☐ Prepare ADNIMERGE2 demo for next class if needed